

Rings and Modules Final (Back Paper) Test

(BMath-2nd year, 2024)

Instructions: Total time 3 Hours. Solve problems (even partially), for a maximum score of 50. Use terminologies, notations and results as covered in the course, no need to prove such results. If you wish to use a problem given in an assignment or homework or any other such, please supply its full solution.

1. Let $f(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{Z}[x]$, $a_n \neq 0$. Suppose that for some k , $0 < k < n$ and some prime p , $p \nmid a_n$, $p \nmid a_k$, $p \mid a_i \forall 0 \leq i \leq k-1$, and $p^2 \nmid a_0$. Show that f has a factor g of degree at least k which is irreducible. (10)
2. Let R be a commutative ring with $1, \neq 0$. Can $R[x, y]$, the ring of polynomials in two variables over R , be a principal ideal ring? Explain. (5)
3. Prove that in a PID, every ideal is a unique product of prime ideals. (5)
4. Let R be a ring and M, N, S be R -modules such that $M \oplus S \cong N \oplus S$. Can we conclude from this that $M \cong N$ as R -modules? Explain. (10)
5. Determine all prime elements in the ring $\mathbb{Z}[i]$ of Gaussian integers. (5)
6. Let k be a field and $A, B \in M_3(k)$. Prove that A is similar to B if and only if $\chi_A = \chi_B$ and $m_A = m_B$. Here χ_A and m_A respectively denote the characteristic and the minimal polynomial of A . Show by an example that these two invariants fail to determine the similarity class of a matrix in general. (10)
7. Let k be a field and $A \in \text{GL}_n(k)$ be such that A is not diagonal and $A^r = I$ for some $r \geq 2$. Give a necessary and sufficient condition on r so that A is diagonalizable, justifying your answer. (10)